05 - Ablation Study

AC-PINN Project | Authors: Suyash Vasal Jain, Nishita Raghvendra

Isolates the contribution of each AC-PINN component on Burgers' equation under noisy sparse data (the hardest condition).

Model	Curriculum	Adaptive Weights
Vanilla PINN	✗	✗
+ Curriculum only	✓	✗
+ Adaptive Weights (ratio) only	✗	ratio
+ Adaptive Weights (gradient) only	✗	gradient
Full AC-PINN (ratio)	✓	ratio
Full AC-PINN (gradient)	✓	gradient
Full AC-PINN (both)	✓	both

In [1]:
```
import sys, os
sys.path.append('..')
import torch
import numpy as np
import matplotlib.pyplot as plt
from pinn_base import (
    device, NoisyDataGenerator, PINNSolver, ACPINNSolver,
    BurgersFDM, Benchmark, save_metrics, save_history
)

PDE        = 'burgers'
LAYERS     = [2, 64, 64, 64, 64, 64, 1]
EPOCHS     = 10000
PDE_PARAMS = {'nu': 0.01/np.pi}
RESULTS    = '../results/burgers/'
FIGURES    = '../figures/burgers/'
os.makedirs(RESULTS, exist_ok=True)
os.makedirs(FIGURES, exist_ok=True)

gen  = NoisyDataGenerator(pde=PDE, **PDE_PARAMS)
# Hardest condition - noisy + sparse
data = gen.generate(N_ic=20, N_bc=20, N_f=2000, noise_eps=0.1)

fdm  = BurgersFDM(nx=256, nt=2000, nu=0.01/np.pi)
fdm.solve()
print(f'Device: {device} | Data ready')
```

```
BurgersFDM solved in 0.6925s
Device: cuda | Data ready
```

Model 1 - Vanilla PINN (no curriculum, no adaptive weights)

In [2]:
```
m1 = PINNSolver(pde=PDE, layers=LAYERS, pde_params=PDE_PARAMS)
h1 = m1.fit(data, epochs=EPOCHS, print_every=2000, label='Ablation | Vanilla')
save_history(h1, RESULTS+'ablation_vanilla.npy')
```

```
D:\PINN\ac-pinn-project\ac-pinn-project\venv\Lib\site-packages\torch\autograd\graph.
py:825: UserWarning: Attempting to run cuBLAS, but there was no current CUDA contex
t! Attempting to set the primary context... (Triggered internally at C:\actions-runn
er\_work\pytorch\pytorch\builder\windows\pytorch\aten\src\ATen\cuda\CublasHandlePoo
l.cpp:135.)
  return Variable._execution_engine.run_backward(  # Calls into the C++ engine to ru
n the backward pass
[Ablation | Vanilla | Epoch     0/10000] Total: 0.918159 | IC: 0.537079 | BC: 0.0944
81 | PDE: 0.057320
[Ablation | Vanilla | Epoch  2000/10000] Total: 0.181021 | IC: 0.072707 | BC: 0.0259
03 | PDE: 0.016482
[Ablation | Vanilla | Epoch  4000/10000] Total: 0.051912 | IC: 0.022000 | BC: 0.0238
96 | PDE: 0.001203
[Ablation | Vanilla | Epoch  6000/10000] Total: 0.039765 | IC: 0.013218 | BC: 0.0226
93 | PDE: 0.000771
[Ablation | Vanilla | Epoch  8000/10000] Total: 0.035702 | IC: 0.010409 | BC: 0.0216
90 | PDE: 0.000721
Training complete in 132.93s
Saved: ../results/burgers/ablation_vanilla.npy
```

Model 2 - Curriculum Only (no adaptive weights)

In [3]:
```
# ACPINNSolver with ratio strategy but effectively fixed weights
# We set weight_update_every very high to disable adaptive updates
m2 = ACPINNSolver(pde=PDE, layers=LAYERS, pde_params=PDE_PARAMS,
                  weight_strategy='ratio')
h2 = m2.fit(data, epochs=EPOCHS, print_every=2000, weight_update_every=999999, labe
save_history(h2, RESULTS+'ablation_curriculum_only.npy')
```

```
[Ablation | Curriculum Only | Epoch     0/10000] Stage 1/4 | Total: 0.518840 | IC:
0.387989 | BC: 0.130283 | PDE: 0.000114 | λ=(1.00,1.00,5.00)
[Ablation | Curriculum Only | Epoch  2000/10000] Stage 1/4 | Total: 0.034756 | IC:
0.010171 | BC: 0.024264 | PDE: 0.000064 | λ=(1.00,1.00,5.00)
[Ablation | Curriculum Only | Epoch  4000/10000] Stage 2/4 | Total: 0.032574 | IC:
0.009539 | BC: 0.022248 | PDE: 0.000157 | λ=(1.00,1.00,5.00)
[Ablation | Curriculum Only | Epoch  6000/10000] Stage 3/4 | Total: 0.036597 | IC:
0.009609 | BC: 0.023557 | PDE: 0.000686 | λ=(1.00,1.00,5.00)
[Ablation | Curriculum Only | Epoch  8000/10000] Stage 4/4 | Total: 0.156823 | IC:
0.098721 | BC: 0.033229 | PDE: 0.004974 | λ=(1.00,1.00,5.00)
AC-PINN training complete in 144.57s
Saved: ../results/burgers/ablation_curriculum_only.npy
```

Model 3 - Adaptive Weights (ratio) Only, no curriculum

In [4]:
```
# Disable curriculum by using very small resample pool
# and always sampling uniformly (stage 4 from epoch 0)
m3 = ACPINNSolver(pde=PDE, layers=LAYERS, pde_params=PDE_PARAMS,
                  weight_strategy='ratio', N_pool=2000, resample_every=1)
# Force stage 4 always by patching sampler thresholds
m3.sampler.STAGE_THRESHOLDS = [0.0, 0.0, 0.0, 1.0]
h3 = m3.fit(data, epochs=EPOCHS, print_every=2000, label='Ablation | Ratio Weights
save_history(h3, RESULTS+'ablation_ratio_weights_only.npy')
```

```
[Ablation | Ratio Weights Only | Epoch     0/10000] Stage 1/4 | Total: 0.934253 | I
C: 0.658402 | BC: 0.065727 | PDE: 0.042025 | λ=(1.00,1.00,5.00)
[Ablation | Ratio Weights Only | Epoch  2000/10000] Stage 4/4 | Total: 0.779098 | I
C: 0.012630 | BC: 0.062665 | PDE: 0.309644 | λ=(0.10,0.49,2.41)
[Ablation | Ratio Weights Only | Epoch  4000/10000] Stage 4/4 | Total: 0.798197 | I
C: 0.011395 | BC: 0.049459 | PDE: 0.309956 | λ=(0.10,0.40,2.51)
[Ablation | Ratio Weights Only | Epoch  6000/10000] Stage 4/4 | Total: 0.596106 | I
C: 0.009877 | BC: 0.031075 | PDE: 0.229523 | λ=(0.11,0.34,2.55)
[Ablation | Ratio Weights Only | Epoch  8000/10000] Stage 4/4 | Total: 0.567800 | I
C: 0.009970 | BC: 0.030682 | PDE: 0.219568 | λ=(0.11,0.35,2.53)
AC-PINN training complete in 170.19s
Saved: ../results/burgers/ablation_ratio_weights_only.npy
```

Model 4 - Adaptive Weights (gradient) Only, no curriculum

In [5]:
```
m4 = ACPINNSolver(pde=PDE, layers=LAYERS, pde_params=PDE_PARAMS,
                  weight_strategy='gradient', N_pool=2000, resample_every=1)
m4.sampler.STAGE_THRESHOLDS = [0.0, 0.0, 0.0, 1.0]
h4 = m4.fit(data, epochs=EPOCHS, print_every=2000, label='Ablation | Gradient Weigh
save_history(h4, RESULTS+'ablation_gradient_weights_only.npy')
```

```
[Ablation | Gradient Weights Only | Epoch     0/10000] Stage 1/4 | Total: 0.844670 |
IC: 0.602279 | BC: 0.049023 | PDE: 0.038674 | λ=(1.00,1.00,5.00)
[Ablation | Gradient Weights Only | Epoch  2000/10000] Stage 4/4 | Total: 0.313442 |
IC: 0.067575 | BC: 0.020971 | PDE: 0.002434 | λ=(1.17,10.00,10.00)
[Ablation | Gradient Weights Only | Epoch  4000/10000] Stage 4/4 | Total: 0.196148 |
IC: 0.028217 | BC: 0.011806 | PDE: 0.008843 | λ=(2.36,9.39,2.13)
[Ablation | Gradient Weights Only | Epoch  6000/10000] Stage 4/4 | Total: 0.133669 |
IC: 0.004546 | BC: 0.007677 | PDE: 0.020619 | λ=(7.64,9.36,1.31)
[Ablation | Gradient Weights Only | Epoch  8000/10000] Stage 4/4 | Total: 0.112024 |
IC: 0.003981 | BC: 0.007653 | PDE: 0.014415 | λ=(8.55,7.69,1.33)
AC-PINN training complete in 170.77s
Saved: ../results/burgers/ablation_gradient_weights_only.npy
```

Model 5 - Full AC-PINN (ratio)

In [6]:
```
m5 = ACPINNSolver(pde=PDE, layers=LAYERS, pde_params=PDE_PARAMS,
                  weight_strategy='ratio', N_pool=20000, resample_every=500)
```

```
h5 = m5.fit(data, epochs=EPOCHS, print_every=2000, label='Ablation | Full AC-PINN R
save_history(h5, RESULTS+'ablation_full_ratio.npy')
```

```
[Ablation | Full AC-PINN Ratio | Epoch     0/10000] Stage 1/4 | Total: 0.564553 | I
C: 0.522554 | BC: 0.037365 | PDE: 0.000927 | λ=(1.00,1.00,5.00)
[Ablation | Full AC-PINN Ratio | Epoch  2000/10000] Stage 1/4 | Total: 0.042353 | I
C: 0.011981 | BC: 0.016175 | PDE: 0.000583 | λ=(1.25,1.69,0.10)
[Ablation | Full AC-PINN Ratio | Epoch  4000/10000] Stage 2/4 | Total: 0.034599 | I
C: 0.010586 | BC: 0.013560 | PDE: 0.001791 | λ=(1.22,1.57,0.21)
[Ablation | Full AC-PINN Ratio | Epoch  6000/10000] Stage 3/4 | Total: 0.024049 | I
C: 0.007757 | BC: 0.009835 | PDE: 0.003570 | λ=(1.10,1.39,0.51)
[Ablation | Full AC-PINN Ratio | Epoch  8000/10000] Stage 4/4 | Total: 0.224630 | I
C: 0.020422 | BC: 0.021768 | PDE: 0.098017 | λ=(0.44,0.47,2.10)
AC-PINN training complete in 144.15s
Saved: ../results/burgers/ablation_full_ratio.npy
```

Model 6 - Full AC-PINN (gradient)

In [7]:

```
m6 = ACPINNSolver(pde=PDE, layers=LAYERS, pde_params=PDE_PARAMS,
                   weight_strategy='gradient', N_pool=20000, resample_every=500)
h6 = m6.fit(data, epochs=EPOCHS, print_every=2000, label='Ablation | Full AC-PINN G
save_history(h6, RESULTS+'ablation_full_gradient.npy')
```

```
[Ablation | Full AC-PINN Gradient | Epoch     0/10000] Stage 1/4 | Total: 0.893116 |
IC: 0.715278 | BC: 0.106101 | PDE: 0.014347 | λ=(1.00,1.00,5.00)
[Ablation | Full AC-PINN Gradient | Epoch  2000/10000] Stage 1/4 | Total: 0.057006 |
IC: 0.009835 | BC: 0.016629 | PDE: 0.000030 | λ=(1.81,2.34,10.00)
[Ablation | Full AC-PINN Gradient | Epoch  4000/10000] Stage 2/4 | Total: 0.037392 |
IC: 0.007327 | BC: 0.009524 | PDE: 0.000035 | λ=(1.92,2.41,10.00)
[Ablation | Full AC-PINN Gradient | Epoch  6000/10000] Stage 3/4 | Total: 0.053895 |
IC: 0.006244 | BC: 0.014439 | PDE: 0.000421 | λ=(4.80,1.36,10.00)
[Ablation | Full AC-PINN Gradient | Epoch  8000/10000] Stage 4/4 | Total: 0.391387 |
IC: 0.069792 | BC: 0.021440 | PDE: 0.020224 | λ=(1.60,10.00,3.24)
AC-PINN training complete in 145.24s
Saved: ../results/burgers/ablation_full_gradient.npy
```

Model 7 - Full AC-PINN (both) - Best Configuration

In [8]:

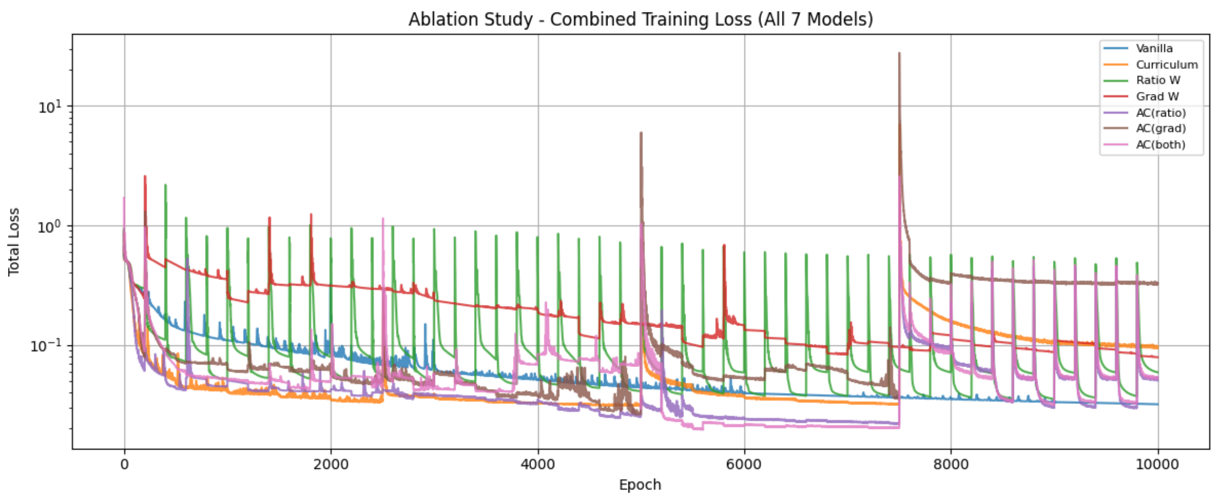
```
m7 = ACPINNSolver(pde=PDE, layers=LAYERS, pde_params=PDE_PARAMS,
                   weight_strategy='both', N_pool=20000, resample_every=500)
h7 = m7.fit(data, epochs=EPOCHS, print_every=2000, label='Ablation | Full AC-PINN B
save_history(h7, RESULTS+'ablation_full_both.npy')
```

```
[Ablation | Full AC-PINN Both | Epoch     0/10000] Stage 1/4 | Total: 1.702283 | IC:
0.510720 | BC: 0.305085 | PDE: 0.177296 | λ=(1.00,1.00,5.00)
[Ablation | Full AC-PINN Both | Epoch  2000/10000] Stage 1/4 | Total: 0.054320 | IC:
0.008353 | BC: 0.011049 | PDE: 0.000003 | λ=(4.81,1.27,10.00)
[Ablation | Full AC-PINN Both | Epoch  4000/10000] Stage 2/4 | Total: 0.083799 | IC:
0.007195 | BC: 0.010164 | PDE: 0.000023 | λ=(10.00,1.14,10.00)
[Ablation | Full AC-PINN Both | Epoch  6000/10000] Stage 3/4 | Total: 0.021524 | IC:
0.006035 | BC: 0.009148 | PDE: 0.002284 | λ=(1.04,1.57,0.39)
[Ablation | Full AC-PINN Both | Epoch  8000/10000] Stage 4/4 | Total: 0.322329 | IC:
0.014929 | BC: 0.023482 | PDE: 0.132707 | λ=(0.26,0.41,2.33)
```

```
AC-PINN training complete in 146.43s
Saved: ../results/burgers/ablation_full_both.npy
```

In [9]:

```
plt.figure(figsize=(12, 5))
for h, lbl in zip([h1, h2, h3, h4, h5, h6, h7],
                  ['Vanilla', 'Curriculum', 'Ratio W', 'Grad W',
                   'AC(ratio)', 'AC(grad)', 'AC(both)']):
    plt.plot(h['total'], label=lbl, alpha=0.8)
plt.yscale('log')
plt.xlabel('Epoch'); plt.ylabel('Total Loss')
plt.title('Ablation Study - Combined Training Loss (All 7 Models)')
plt.legend(fontsize=8); plt.grid(True)
plt.tight_layout()
plt.savefig(FIGURES+'ablation_training_all.png', dpi=150)
plt.show()
```



Ablation Benchmark Table

In [10]:

```
models = [
    ('Vanilla PINN',                    m1),
    ('+ Curriculum only',               m2),
    ('+ Ratio weights only',            m3),
    ('+ Gradient weights only',         m4),
    ('Full AC-PINN (ratio)',            m5),
    ('Full AC-PINN (gradient)',         m6),
    ('Full AC-PINN (both) ← Best',     m7),
]

bench = Benchmark(fdm, nx=200, nt=100)
for name, model in models:
    bench.add(name, model)
bench.run()

ablation_metrics = bench.compare_metrics()
save_metrics(ablation_metrics, RESULTS+'ablation_metrics.npy')
```

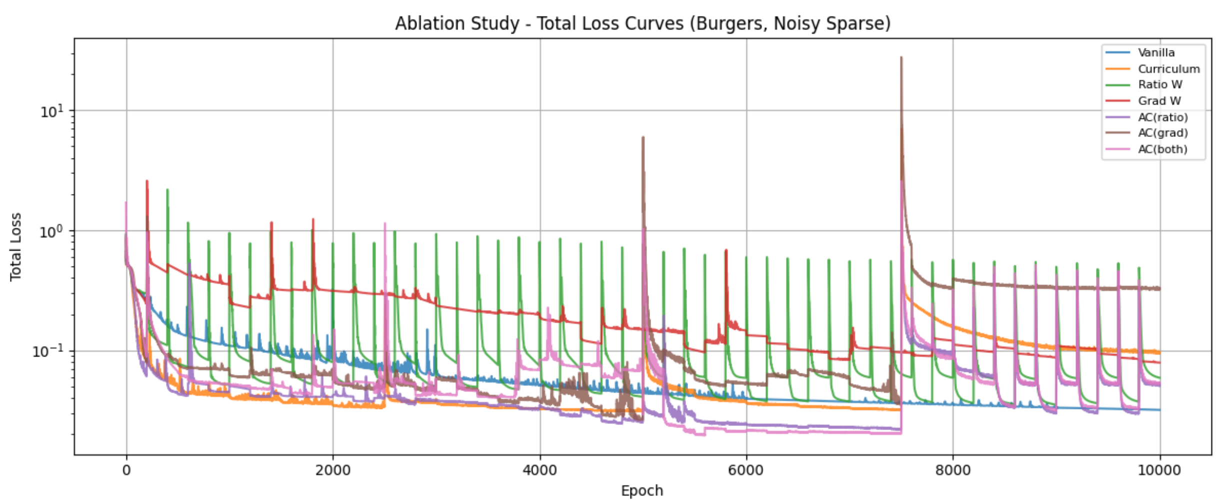
```
=================================================================
  Model                          Rel L2    Max Err       MAE       RMSE
=================================================================
  Vanilla PINN                 0.521381   1.775960   0.159925   0.317085
  + Curriculum only            0.560341   1.450891   0.191445   0.340780
  + Ratio weights only         0.396582   1.039698   0.133330   0.241187
  + Gradient weights only      0.570826   1.853476   0.160891   0.347156
  Full AC-PINN (ratio)         0.354727   0.943174   0.121277   0.215732
  Full AC-PINN (gradient)      0.581307   1.340159   0.206489   0.353530
  Full AC-PINN (both) ← Best   0.358250   0.994273   0.128044   0.217875
=================================================================
Saved: ../results/burgers/ablation_metrics.npy
```

Ablation Loss Curves

In [11]:

```python
histories = [h1, h2, h3, h4, h5, h6, h7]
labels    = ['Vanilla', 'Curriculum', 'Ratio W', 'Grad W',
             'AC(ratio)', 'AC(grad)', 'AC(both)']

plt.figure(figsize=(12, 5))
for h, lbl in zip(histories, labels):
    plt.plot(h['total'], label=lbl, alpha=0.8)
plt.yscale('log')
plt.xlabel('Epoch'); plt.ylabel('Total Loss')
plt.title('Ablation Study - Total Loss Curves (Burgers, Noisy Sparse)')
plt.legend(fontsize=8); plt.grid(True)
plt.tight_layout()
plt.savefig(FIGURES+'ablation_loss_curves.png', dpi=150)
plt.show()
print('Ablation study complete.')
```




```
Ablation study complete.
```